

# **Lutein and zeaxanthin intake and the risk of age-related macular degeneration: a systematic review and meta-analysis.**



Lutein and zeaxanthin are thought to decrease the incidence of age-related macular degeneration (AMD); however, findings have been inconsistent. We conducted a systematic literature review and meta-analysis to evaluate the relationship between dietary intake of lutein and zeaxanthin and AMD risk. Relevant studies were identified by searching five databases up to April 2010. Reference lists of articles were retrieved, and experts were contacted. Literature search, data extraction and study quality assessment were performed independently by two reviewers and results were pooled quantitatively using meta-analysis methods. The potential sources of heterogeneity and publication bias were also estimated. The search yielded six longitudinal cohort studies. The pooled relative risk (RR) for early AMD, comparing the highest with the lowest category of lutein and zeaxanthin intake, was 0·96 (95 % CI 0·78, 1·17). Dietary intake of these carotenoids was significantly related with a reduction in risk of late AMD (RR 0·74; 95 % CI 0·57, 0·97); and a statistically significant inverse association was observed between lutein and zeaxanthin intake and neovascular AMD risk (RR 0·68; 95 % CI 0·51, 0·92). The results were essentially consistent among subgroups stratified by participant characteristics. The findings of the present meta-analysis indicate that dietary lutein and zeaxanthin is not significantly associated with a reduced risk of early AMD, whereas an increase in the intake of these carotenoids may be protective against late AMD. However, additional studies are needed to confirm these relationships.